

Swiggy's Restaurant Recommendation system



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A machine learning-based recommendation system that suggests restaurants to users based on their preferences, location, and dining history using Streamlit.



Project Overview

Methodology:

Data Processing

Data Cleaning: Handle missing values, remove duplicates

Encoding: One-Hot Encoding for categorical features

Clustering: K-Means clustering for grouping similar restaurants

Recommendation Algorithms

Cluster-based: Find restaurants in the same cluster

Preference-based: Filter by user criteria

Similarity-based: Cosine similarity between feature vectors



Project Overview

Features:

Three Recommendation Methods:

-  Cluster-based recommendations
-  Preference-based filtering
-  Cosine similarity matching

Smart Filtering Options:

-  City selection
-  Cuisine preferences
-  Minimum rating
-  Maximum budget

User-Friendly Interface:

-  Clean Streamlit interface
-  Responsive design
-  Real-time results



Technologies Used

Frontend: Streamlit

Backend: Python, Pandas, NumPy

ML: Scikit-learn (K-Means, Cosine Similarity)



Run a code

1. Data cleaning impute

`python scripts/data_cleaning_impute.py`

2. Data Cleaning drop

`python scripts/data_cleaning_drop.py`

3. Data Processing

`python scripts/data_processing.py`

4. Indices

`python scripts/indices.py`

5. Recommendation

`python scripts/recommendation.py`

6. Streamlit

`python scripts/stream.py`



Filter Preferences

City

Porur,Chennai

Cuisine Types

Biryani

Beverages

Sweets,Bakery

Fast Food,Indian

Minimum Rating

3.00

0.00

5.00

Price Range

174700

0

300450

Recommendation Method

Similarity-Based

Cluster-Based

 Restaurant Recommendation Engine

Get Recommendations

Top Recommended Restaurants in Porur,Chennai

Restaurant	City	Cuisine	Rating	Cost
THIRSTY REMEDY	Porur,Chennai	Beverages	4.0 	₹200
LASSI BLAST	Porur,Chennai	Beverages	4.0 	₹200
Nayaab Biryani	Porur,Chennai	Biryani	4.0 	₹350
SAY CHEESE VALASARAVAKKAM	Porur,Chennai	Beverages	4.0 	₹250
Bicycle Thieves Cafe	Porur,Chennai	Beverages	4.0 	₹175
THE WRAP STORY	Porur,Chennai	Beverages	4.0 	₹200
GURU TEA STALL & TIFFIN CENTRE	Porur,Chennai	Beverages	4.0 	₹300
Lassi Nation	Porur,Chennai	Beverages	4.0 	₹200
Mummys briyani	Porur,Chennai	Biryani	3.8 	₹150



Filter Preferences

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Porur,Chennai

Cuisine Types

Biryani

Minimum Rating

0.003.505.00

Price Range

03003500300450

Recommendation Method

☐ Similarity-Based

☒ Cluster-Based



Restaurant Recommendation Engine

Get Recommendations

Top Recommended Restaurants in Porur,Chennai

Restaurant	City	Cuisine	Rating	Cost
Musthar Briyani	Porur,Chennai	Biryani	4.1 ★	₹300
Jallikattu Biriyani	Porur,Chennai	Biryani	4.0 ★	₹400
HS BIRYANI	Porur,Chennai	Biryani	4.0 ★	₹300
Nayaab Biryani	Porur,Chennai	Biryani	4.0 ★	₹350
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